

BOOLEAN FUNCTIONS & BOOLEAN LAWS

Unit 2 - Digital Electronics

DIGITS USED IN NUMBER SYSTEMS

Number System	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Binary															1	0
Octal									7	6	5	4	3	2	1	0
Decimal							9	8	7	6	5	4	3	2	1	0
Hexadecimal	F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0

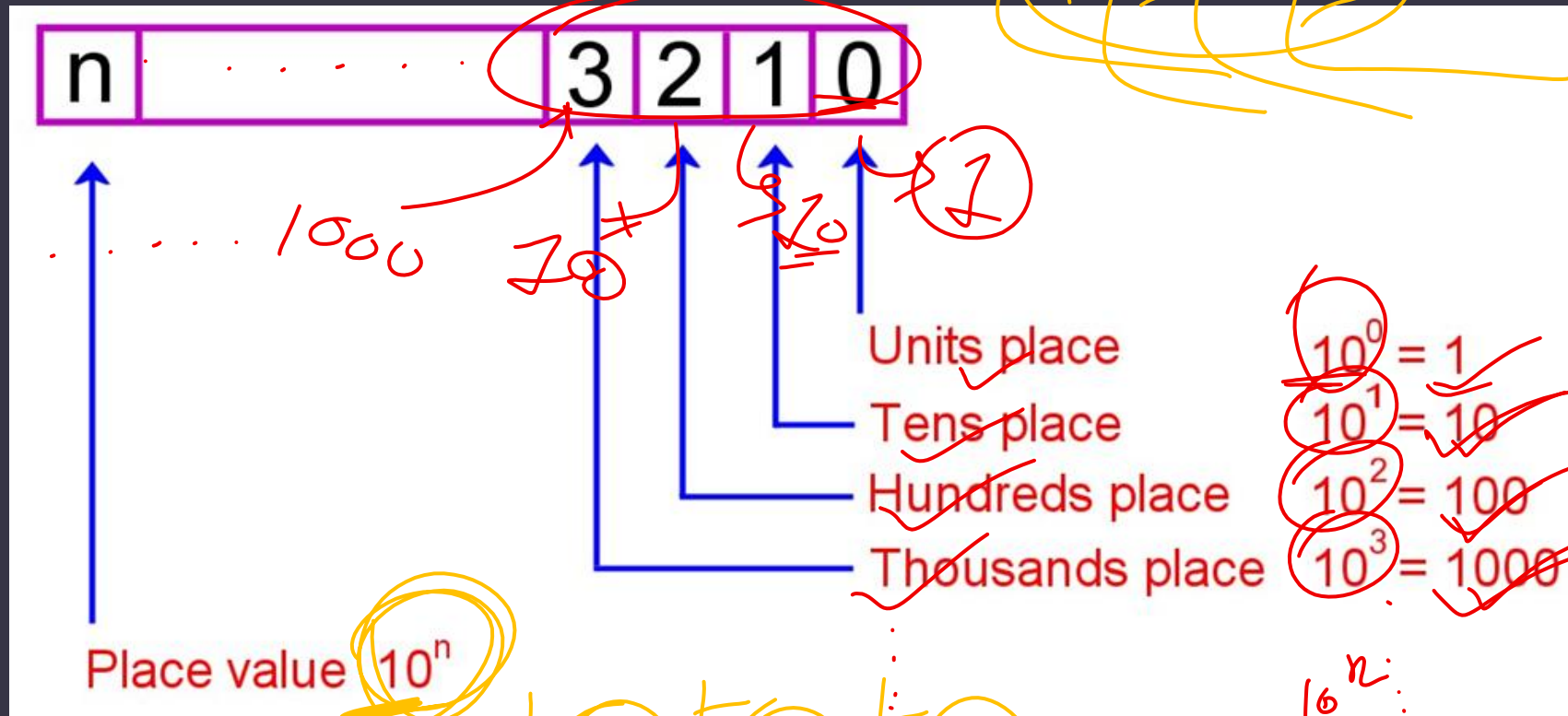
Base $\Rightarrow 2$

Base $\Rightarrow 8$

Base $\Rightarrow 10$

Base $\Rightarrow 16$

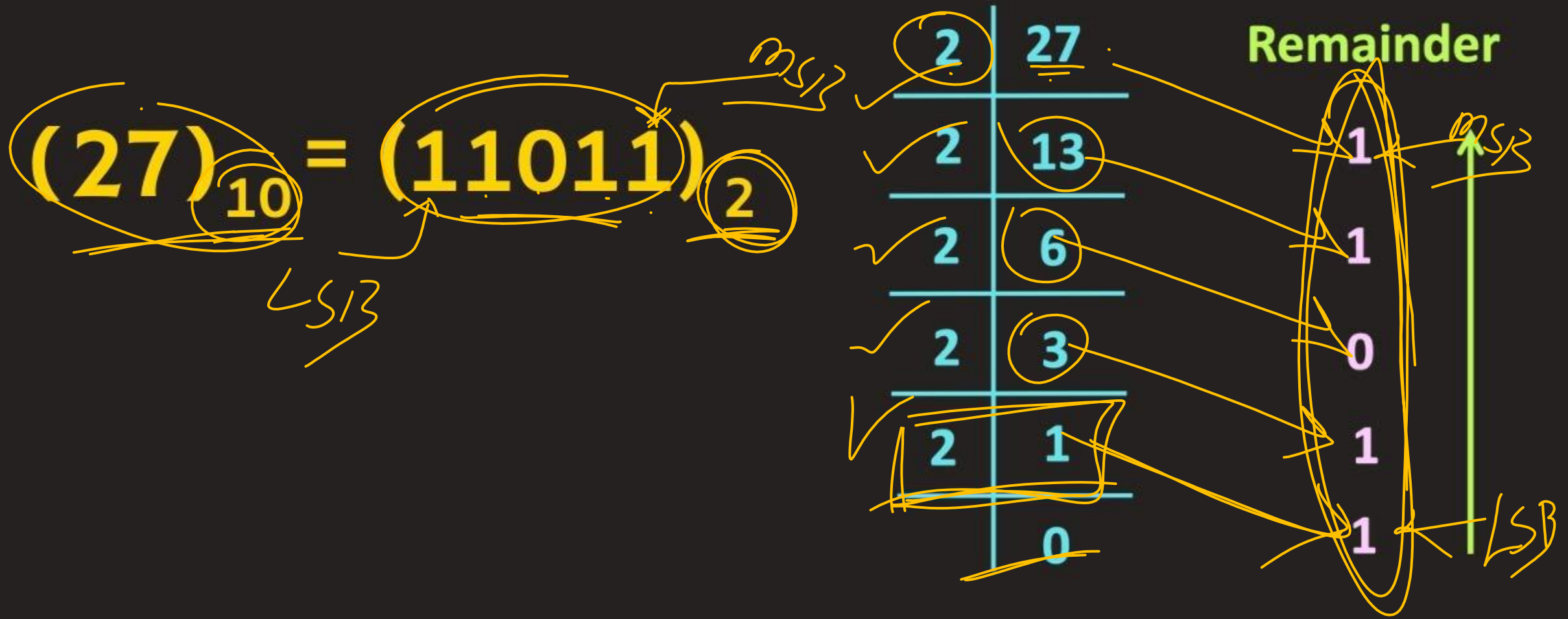
DECIMAL NUMBERING SYSTEM



Handwritten calculation showing the expansion of the number 3210:

$$3 \times 1000 + 2 \times 100 + 1 \times 10 + 0 \times 1 = 3000 + 200 + 10 + 0$$

Decimal to Binary Conversion



Decimal to Binary

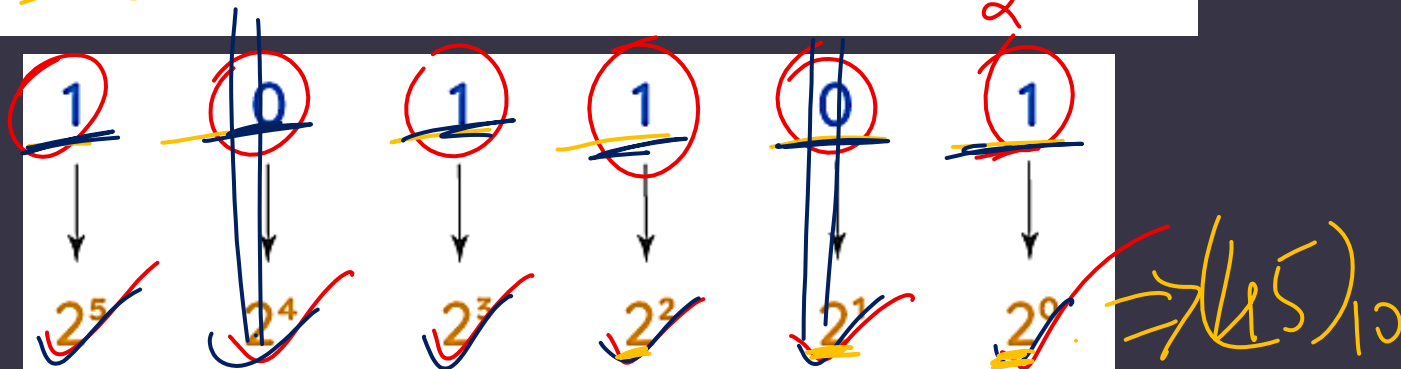
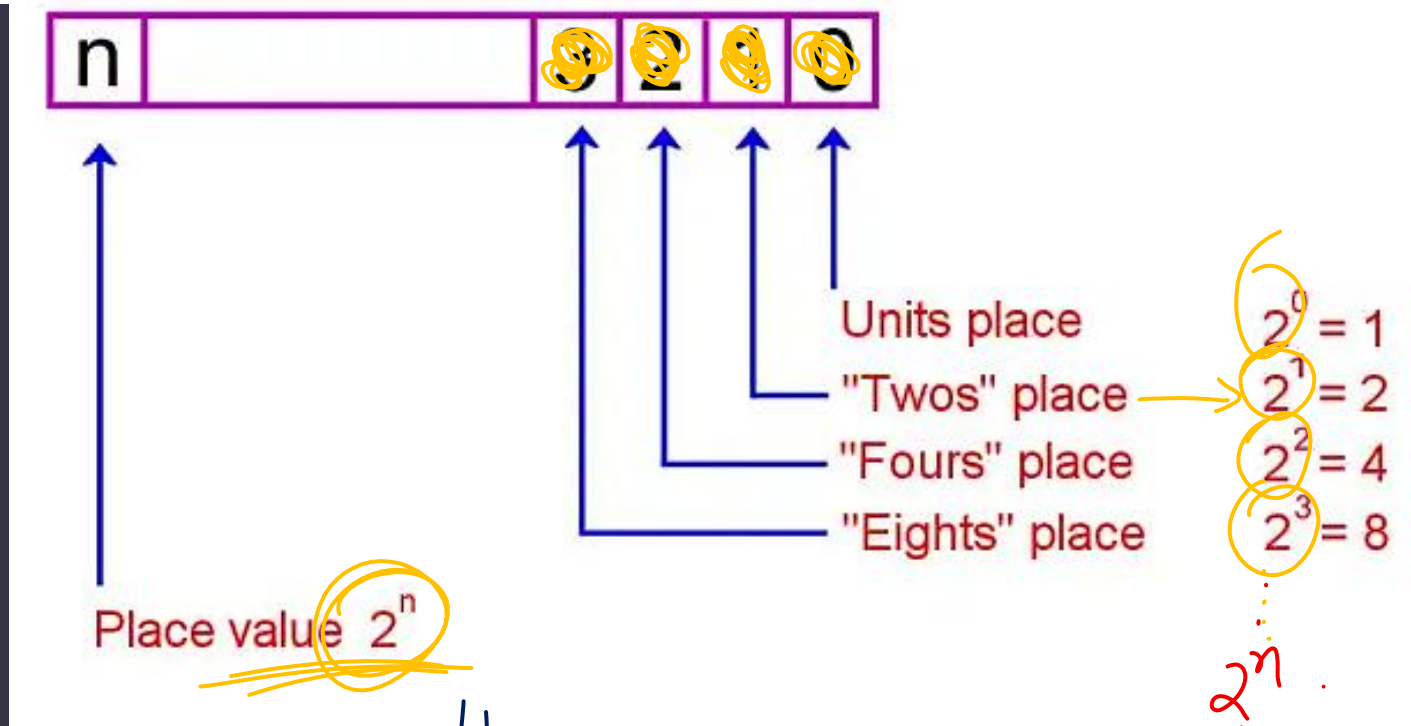
47	÷	2	=	<u>23</u>	Remainder 1	
23	÷	2	=	11	Remainder 1	
11	÷	2	=	5	Remainder 1	
5	÷	2	=	2	Remainder 1	
2	÷	2	=	1	Remainder 0	
1	÷	2	=	0	Remainder 1	

1 0 1 1 1 1

Divide by 2 stops
as quotient reaches 0

$$\underline{(47)}_{10} = \underline{(101111)}_2$$

BINARY NUMBERING SYSTEM



BINARY TO DECIMAL

1 1 0 1 1 0 1 1

Binary to Decimal

$$\begin{aligned} 1 \times 2^0 &= 1 \times 1 = 1 \\ 1 \times 2^1 &= 1 \times 2 = 2 \\ 0 \times 2^2 &= 0 \times 4 = 0 \\ 1 \times 2^3 &= 1 \times 8 = 8 \\ 1 \times 2^4 &= 1 \times 16 = 16 \\ 0 \times 2^5 &= 0 \times 32 = 0 \\ 1 \times 2^6 &= 1 \times 64 = 64 \\ 1 \times 2^7 &= 1 \times 128 = 128 \end{aligned}$$

$$1 + 2 + 8 + 16 + 64 + 128 = 219$$

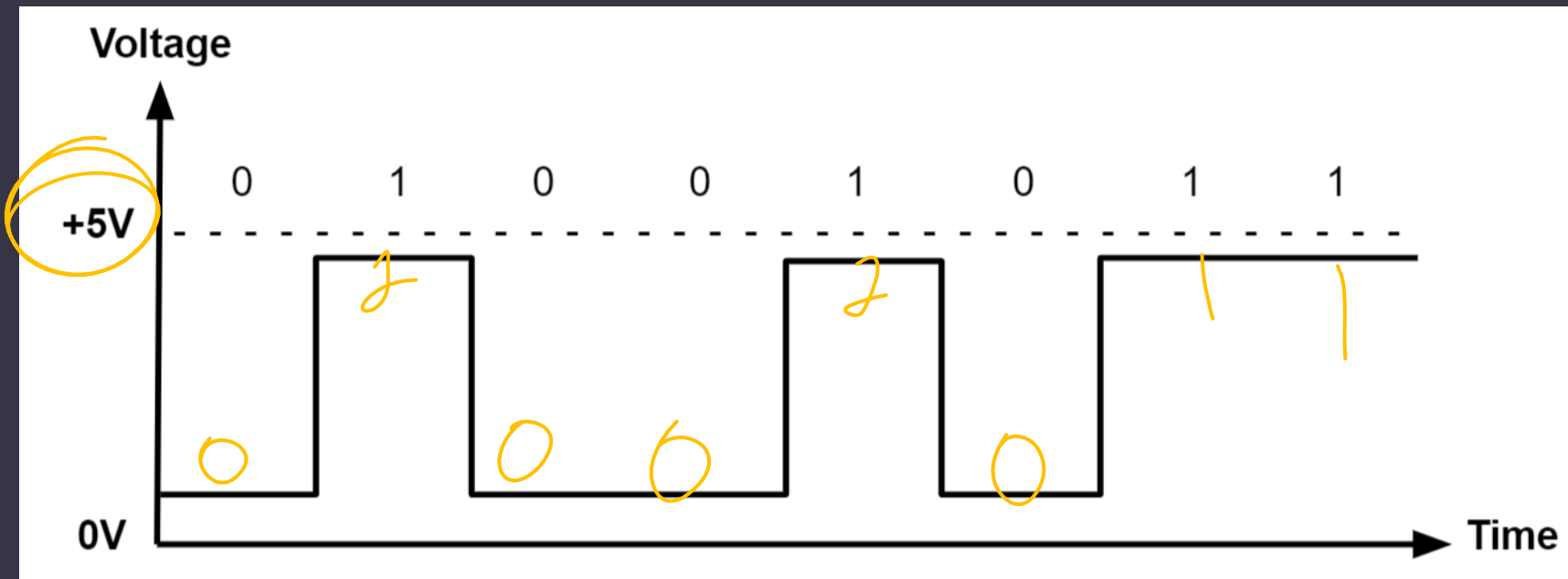
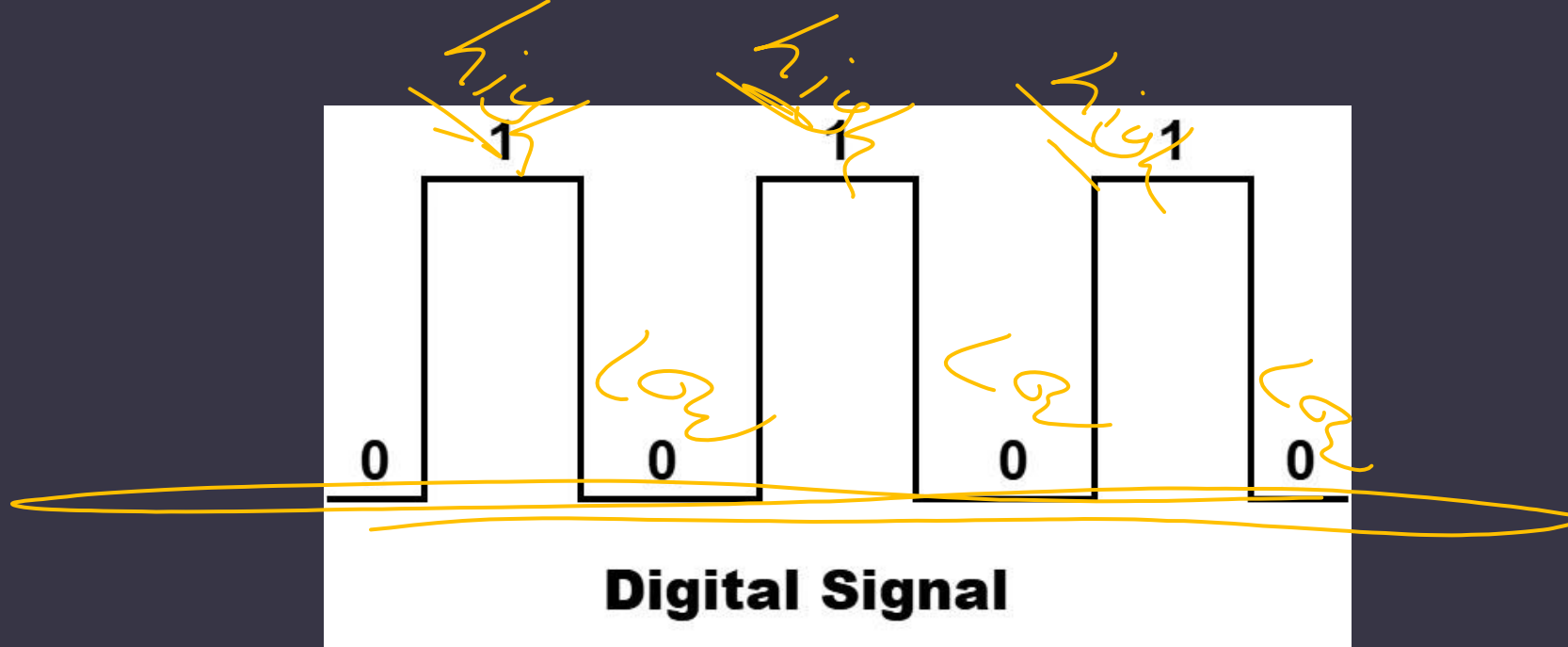
$$(11011011)_2 = (219)_{10}$$

$$\begin{array}{r} 256 \\ + 128 \\ + 32 \\ + 4 \\ \hline \end{array}$$

$$(110100100)_2$$

DIGITAL SIGNAL

- An analog signal is a continuous time-varying current or voltage signal,
- whereas a digital signal is a pulsating waveform of two discrete values: *high* and *low*.
- These two discrete values are represented by binary numbers 0 and 1.
- A digital circuit is an electronic circuit that processes digital signals.
- In digital circuits,
 - 0 = No, False, Switch Off, Open Circuit, and Low.
 - 1 = Yes, True, Switch On, Closed Circuit, and High.



WHAT IS BOOLEAN LOGIC?

high 1
low 0

- Boolean logic is a type of algebra in which results are calculated as either TRUE or FALSE (known as truth values or truth variables).
- Boolean logic utilizes three basic logical operators: AND, OR, and NOT.
- In other words, Boolean means using operators to compare variables against the criteria you define to determine if the variables meet them.
- A Boolean operation, then, is the construction of a logical expression to help carry out that function.

Boolean expression
SOP (minterms)
POS (maxterms)

BOOLEAN OPERATORS

- **AND:** The Boolean AND operator is used to confirm that two or more Boolean expressions are all true.
- **OR:** The Boolean OR operator checks that either one condition or another is true.
- **NOT:** takes only one ^{IF} argument and changes TRUE value to FALSE and FALSE value to TRUE.

→ Inverted

LOGIC GATES

- A logic gate is a digital circuit with a single output whose value depends upon the logical relationship between the input(s) and output.
- In simple words, The relationship between the input values and the output is based on a certain 'logic', hence these circuits are addressed as logic gates.
- The logic gates have only one output for a single input or multiple inputs.

TYPES OF LOGIC GATES

There are 3 types of logic gates:

1) **Basic Gates:** OR, AND, and NOT Gates.

2) **Universal Gates:** NAND, and NOR Gates.

3) **Derived Gates:** XOR Gates, and XNOR Gates.

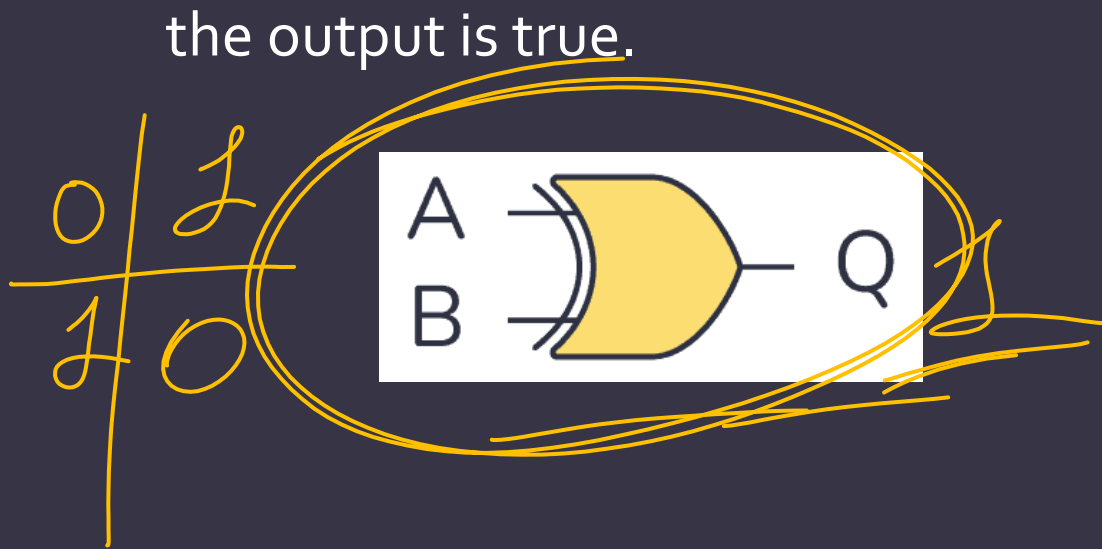
Handwritten notes in yellow ink:

- An arrow points from "NOT" in item 1 to "NOT" in item 2.
- An arrow points from "AND" in item 1 to "AND" in item 2.
- An arrow points from "XOR" in item 3 to "NOT" in item 2.
- An arrow points from "XNOR" in item 3 to "OR" in item 2.

NOT AND
NOT OR

EXCLUSIVE OR, EX-OR OR XOR GATE

- The XOR gate (or Exclusive OR) outputs 1 if one of its two inputs is 1 – but not both. $Q = A \text{ XOR } B = (A \oplus B) = A'B + AB'$
- You can also look at it this way – if the two inputs are different from each other, the output is true.



Input A	Input B	Output Q	Q
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

EXCLUSIVE NOR, EX-NOR, OR XNOR GATE

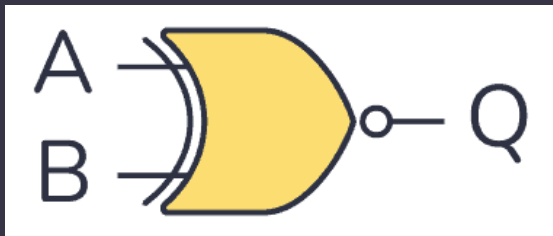
- EX-NOR gate has two or more inputs but a single output. $Q = A \text{ XNOR } B = A \odot B$

$$= AB + A'B'$$



- EX-NOR gate generates a HIGH (1) output if both the inputs are at the same logic level $A=B$.

0	1
0	1



$\Rightarrow 2$

Input A	Input B	Output Q
0	0	1
0	1	0
1	0	0
1	1	1

BOOLEAN EXPRESSIONS

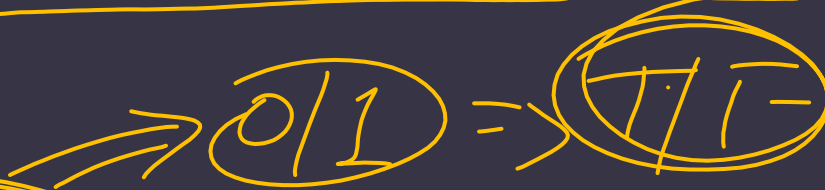
→ minterms (SOP)
→ Maxterms (POS)

- An Irish mathematician 'George Boole' developed a mathematical system with the use of symbols.
- Boolean expression (also known as logical expression) is an expression used that produces a Boolean value when evaluated.
- A Boolean value is either true or false.
- A Boolean expression may be composed of a combination of the Boolean constants True/False, Boolean-typed variables, Boolean-valued operators, and Boolean-valued functions.
- $A + B$, $A'B'C' + A'BC + ABC'$, etc ...

$$A'B' \oplus AB, A'B + A'B'$$

BOOLEAN FUNCTIONS

- A Boolean function is a mathematical function that maps arguments to a value,



↳ where the allowable values of range (the function arguments) and

↳ domain (the function value) are just one of two values:

➤ true and false (or 0 and 1).

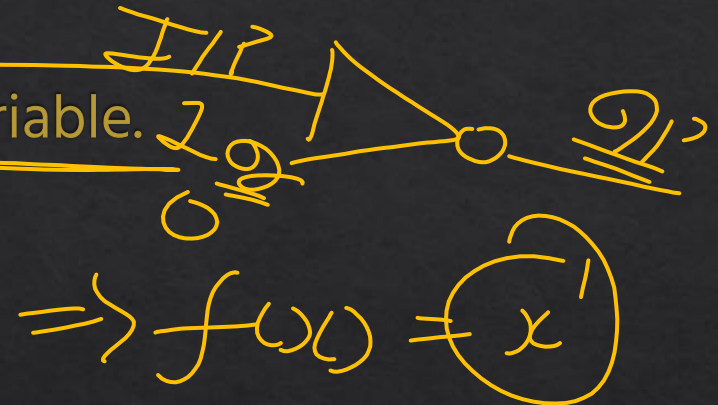
- The study of Boolean functions is known as Boolean logic.

Define Boolean Function

- ◇ To define any Boolean function, we need only to specify its value for each possible value of its inputs.

- ◇ The not function is a Boolean function of one variable.

$$\underline{NOT(x)} = \begin{cases} 1 & \text{if } x \text{ is } 0 \\ 0 & \text{if } x \text{ is } 1 \end{cases}$$



- ◇ The and, or, and exclusive or functions are familiar Boolean functions of two variables.

$$\underline{AND(x, y)} = \begin{cases} 1 & \text{if both } x \text{ and } y \text{ are } 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\Rightarrow f(x, y) = A \cdot B$$

$$\underline{OR(x, y)} = \begin{cases} 1 & \text{if either } x \text{ or } y \text{ (or both) is } 1 \\ 0 & \text{otherwise} \end{cases}$$

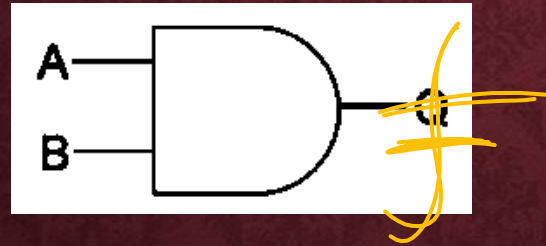
$$\Rightarrow f(x, y) = A + B$$

$$\underline{XOR(x, y)} = \begin{cases} 1 & \text{if } x \text{ and } y \text{ are different} \\ 0 & \text{otherwise} \end{cases}$$

$$\Rightarrow f(x, y) = A'B + AB'$$

- So, Boolean Function looks like this: $F(x, y, \dots) = x + y' + xy \dots$
- where $x, y, \dots$ are the Variables/Inputs and F is the output that comes when putting those Inputs into the Boolean Function F .
- For example, an AND Gate can be represented as a Boolean Function like that:

$$Q(A, B) = A \cdot B$$



- Each row of the Truth Table represents an minterm or maxterm.
- In Short, if we have input Variables and we want to represent it as Boolean Function:
- We can write it in either using minterms or using maxterms.

BOOLEAN ALGEBRA AXIOMS (POSTULATES)

An axiom is nothing more than the definition of three basic logic operations (AND, OR and NOT).

All axioms defined in Boolean algebra are the results of an operation that is performed by a logical gate.

$$\cdot 0 \cdot 0 = 0$$

$$\cdot 0 + 0 = 0$$

$$\cdot 1 \cdot 1 = 1$$

$$\cdot 1 + 1 = 1$$

$$\cdot 0 \cdot 1 = 1 \cdot 0 = 0$$

$$\cdot 0 + 1 = 1 + 0 = 1$$

$$\cdot 0' = 1$$

$$\cdot 1' = 0$$